ANÁLISE DE DADOS END-TO-END

Telco Customer Churn:

Uma mesma base de dados sob três perspectivas analíticas para identificar gargalos operacionais.

7.043
Total de Clientes

26,54%
Taxa Geral de Churn

3 Ferramentas
Excel • SQL • Power BI

Pergunta de Negócio:

“A empresa quer reduzir o cancelamento (churn) de clientes. Quais características de contrato e forma de pagamento têm a maior taxa de churn, e qual segmento de clientes deveria receber uma ação de retenção prioritária?"

ETAPA 01

Excel:

Rótulos de Linha	Contagem de Churn
No	73,46%
Yes	26,54%
Total Geral	**100,00%**

Contagem de custor	Rótulos de Coluna		
Rótulos de Linha	No	Yes	Total Geral
Month-to-month	57,29%	42,71%	100,00%
One year	88,73%	11,27%	100,00%
Two year	97,17%	2,83%	100,00%
Total Geral	**73,46%**	**26,54%**	**100,00%**

Contagem de custor	Rótulos de Coluna		
Rótulos de Linha	No	Yes	Total Geral
0 - 10	50,9%	49,1%	100,0%
10 - 20	68,8%	31,2%	100,0%
20 - 30	77,2%	22,8%	100,0%
30 - 40	78,1%	21,9%	100,0%
40 - 50	82,4%	17,6%	100,0%
50 - 60	86,4%	13,6%	100,0%
60 - 70	90,7%	9,3%	100,0%
70 - 80	97,7%	2,3%	100,0%
Total Geral	**73,5%**	**26,5%**	**100,0%**

Contagem de customerID	Rótulos de Coluna		
Rótulos de Linha	No	Yes	Total Geral
Month-to-month	**57%**	**43%**	**100%**
Bank transfer (automatic)	66%	34%	100%
Credit card (automatic)	67%	33%	100%
Electronic check	46%	54%	100%
Mailed check	68%	32%	100%
One year	**89%**	**11%**	**100%**
Bank transfer (automatic)	90%	10%	100%
Credit card (automatic)	90%	10%	100%
Electronic check	82%	18%	100%
Mailed check	93%	7%	100%
Two year	**97%**	**3%**	**100%**
Bank transfer (automatic)	97%	3%	100%
Credit card (automatic)	98%	2%	100%
Electronic check	92%	8%	100%
Mailed check	99%	1%	100%
Total Geral	**73%**	**27%**	**100%**

ETAPA 02

Análise de Permanência (Tenure) em SQL

Uso de WITH (CTE) para categorizar os meses de contrato sem alterar a tabela original no banco de dados.

Agrupamento por Faixas (Bucket)

```sql
WITH TabelaComBucket AS (
  SELECT customerID, Churn,
    CASE
      WHEN CAST(tenure AS INT) BETWEEN 0 AND 10 THEN '0 - 10'
      WHEN CAST(tenure AS INT) BETWEEN 11 AND 20 THEN '10 - 20'
      WHEN CAST(tenure AS INT) BETWEEN 21 AND 30 THEN '20 - 30'
      WHEN CAST(tenure AS INT) BETWEEN 31 AND 40 THEN '30 - 40'
      WHEN CAST(tenure AS INT) BETWEEN 41 AND 50 THEN '40 - 50'
      WHEN CAST(tenure AS INT) BETWEEN 51 AND 60 THEN '50 - 60'
      WHEN CAST(tenure AS INT) BETWEEN 61 AND 70 THEN '60 - 70'
      ELSE '70 - 80'
    END AS Bucket_Tenure
  FROM telefonia
)
SELECT Bucket_Tenure, Churn, COUNT(*) AS Qtd
FROM TabelaComBucket
    GROUP BY Bucket_Tenure, Churn;
```

```sql
WITH TabelaComBucket AS (
    SELECT
        customerID,
        Churn,
        CASE
            WHEN CAST(tenure AS INT) BETWEEN 0 AND 10 THEN '0 - 10'
            WHEN CAST(tenure AS INT) BETWEEN 11 AND 20 THEN '10 - 20'
            WHEN CAST(tenure AS INT) BETWEEN 21 AND 30 THEN '20 - 30'
            WHEN CAST(tenure AS INT) BETWEEN 31 AND 40 THEN '30 - 40'
            WHEN CAST(tenure AS INT) BETWEEN 41 AND 50 THEN '40 - 50'
            WHEN CAST(tenure AS INT) BETWEEN 51 AND 60 THEN '50 - 60'
            WHEN CAST(tenure AS INT) BETWEEN 61 AND 70 THEN '60 - 70'
            ELSE '70 - 80'
        END AS Bucket_Tenure
FROM telefonia)
SELECT
    Bucket_Tenure,
    Churn,
    COUNT(customerID) AS 'Qtd Clientes',
    ROUND(
        COUNT(customerID) * 100.0 / SUM(COUNT(customerID)) OVER(PARTITION BY Bucket_Tenure),
        1
    ) || '%' AS 'Percentual por Bucket'
FROM
    TabelaComBucket
GROUP BY
    Bucket_Tenure, Churn
ORDER BY
    Bucket_Tenure, Churn;
```

Exibição em grade | Visualização do formulário

Total de linhas carregadas: 16

	Bucket_Tenure	Churn	Qtd Clientes	Percentual por Bucket
1	0 - 10	No	1002	50.9%
2	0 - 10	Yes	968	49.1%
3	10 - 20	No	625	68.8%
4	10 - 20	Yes	283	31.2%
5	20 - 30	No	589	77.2%
6	20 - 30	Yes	174	22.8%
7	30 - 40	No	504	78.1%
8	30 - 40	Yes	141	21.9%
9	40 - 50	No	537	82.4%
10	40 - 50	Yes	115	17.6%
11	50 - 60	No	603	86.4%
12	50 - 60	Yes	95	13.6%
13	60 - 70	No	794	90.7%
14	60 - 70	Yes	81	9.3%
15	70 - 80	No	520	97.7%
16	70 - 80	Yes	12	2.3%

ETAPA 03

Cruzamento de Contrato & Pagamento

Identificação do Gargalo Crítico

O cruzamento entre o tipo de contrato e o método de pagamento revelou uma concentração expressiva de cancelamentos:

- Month-to-month + Electronic check: Alcança 54% de Churn.
- Contratos de 2 anos: Taxa média de Churn despenca para apenas 2.83%.

```sql
SELECT
    Contract,
    PaymentMethod,
    Churn,
    COUNT(customerID) AS 'QTD Cliente',
    ROUND(
        COUNT(customerID) * 100.0 / SUM(COUNT(customerID)) OVER(PARTITION BY Contract, PaymentMethod),0) || '%' AS '% por Contrato e Pagamento'
FROM
    telefonia
GROUP BY
    Contract,
    PaymentMethod,
    Churn
ORDER BY
    Contract,
    PaymentMethod,
    Churn;
```

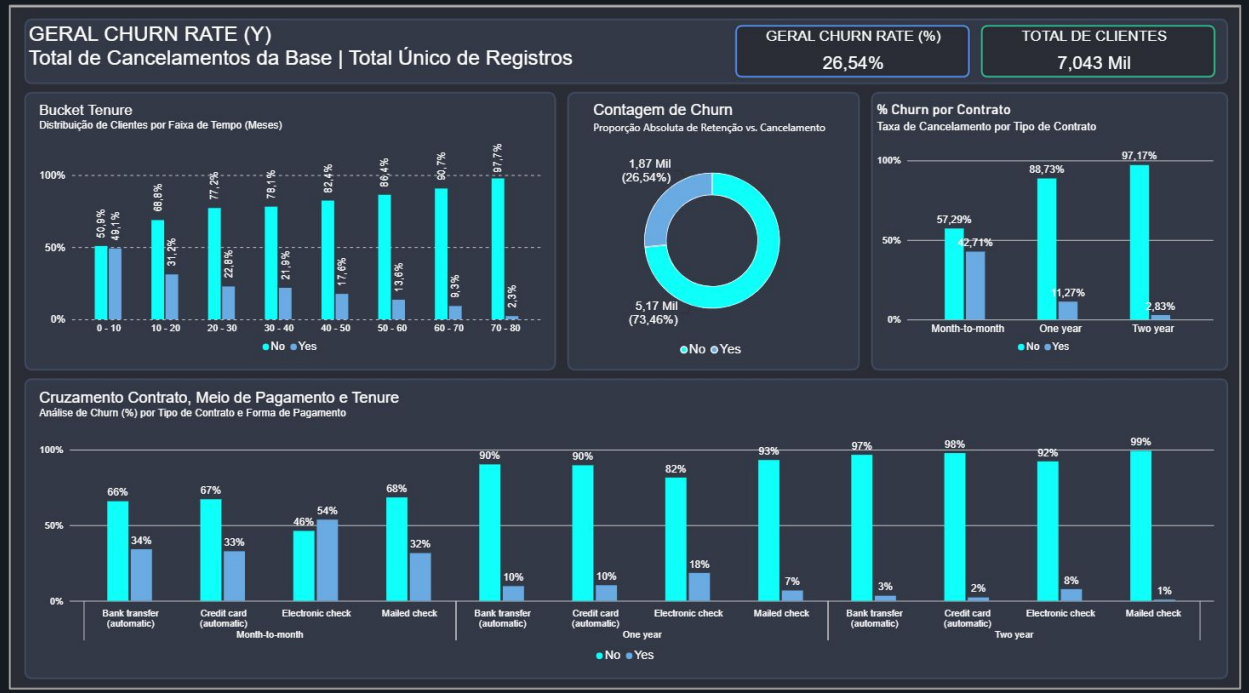
Exibição em grade | Visualização do formulário

Total de linhas carregadas: 24

	Contract	PaymentMethod	Churn	QTD Cliente	% por Contrato e Pagamento
1	Month-to-month	Bank transfer (automatic)	No	388	66.0%
2	Month-to-month	Bank transfer (automatic)	Yes	201	34.0%
3	Month-to-month	Credit card (automatic)	No	365	67.0%
4	Month-to-month	Credit card (automatic)	Yes	178	33.0%
5	Month-to-month	Electronic check	No	856	46.0%
6	Month-to-month	Electronic check	Yes	994	54.0%
7	Month-to-month	Mailed check	No	611	68.0%
8	Month-to-month	Mailed check	Yes	282	32.0%
9	One year	Bank transfer (automatic)	No	353	90.0%
10	One year	Bank transfer (automatic)	Yes	38	10.0%
11	One year	Credit card (automatic)	No	357	90.0%
12	One year	Credit card (automatic)	Yes	41	10.0%
13	One year	Electronic check	No	283	82.0%
14	One year	Electronic check	Yes	64	18.0%
15	One year	Mailed check	No	314	93.0%
16	One year	Mailed check	Yes	23	7.0%
17	Two year	Bank transfer (automatic)	No	545	97.0%
18	Two year	Bank transfer (automatic)	Yes	19	3.0%
19	Two year	Credit card (automatic)	No	568	98.0%
20	Two year	Credit card (automatic)	Yes	13	2.0%
21	Two year	Electronic check	No	155	92.0%
22	Two year	Electronic check	Yes	13	8.0%
23	Two year	Mailed check	No	379	99.0%
24	Two year	Mailed check	Yes	3	1.0%

ETAPA 04

Dashboard Interativo no Power BI



CONCLUSÃO

Insights & Plano de Ação

Dimensão	Gargalo Identificado	Ação Recomendada
Tenure (Tempo)	49.1% de Churn nos primeiros 10 meses de contrato.	Programa de Onboarding e acompanhamento proativo nos primeiros 90 dias.
Tipo de Contrato	Contratos mensais (Month-to-month) registram 42.7% de cancelamento.	Incentivo financeiro/desconto para migração para planos anuais.
Forma de Pagamento	Electronic check apresenta a maior taxa de saída (54%).	Campanha para adesão ao Débito Automático ou Cartão de Crédito.